

# US Airforce IFO-21

Flight model: Boeing 737  
Date: April 03, 1996

From: Zagreb, Croatia  
To: Dubronik, Croatia

The plane carrying high level military officers and US Secretary of commerce of state. The plane have a stop over at Tuzla, Bosnia. Bosnia is in middle of a civil war. The weather is very rainy and filled with fog.

## Problem

The plane had to change its course due to airspace closure. Due to the civil war, all the instrumental landing system in the Dubronik airport which include **Doppler radar** warns of incoming storms, glide slope which tells the pilots on the glide angle and even radar that tells the ATC of incoming flights to the airport instead uses **non directional beacons** towards the run to guide planes towards runway which is an old technology. The airport here have to beacons when the signal from the first beacon is heard the crew must refer to their charts and should follows the mentioned directions. The clouds are very thick and the crew cannot see the ground. The crew keeps looking for the airport but finds a mountain. Despite attempts to pull up they crash into the mountains. The plane had a **CPI (Crash position indicator)** which sends high frequency signal in case of any crashes but the same airport do not have the equipment to pickup the signal.

## Investigation

The plane had no CVR or FDR as the planes were a part of the airforce. The plane had crashed 3 km away from the airport. The plane deviates from the part from last 20 km. The plane was on tract all throughout its journey and suffered no deviation except the above mentioned. The investigators finds **ADF (Automatic direction finder)** Which guides the plane from the signal received. As it had only one beacon instead of 2 it can either of the one beacon and not both at the same time. The investigators says that the crew amid the chaos relayed on even older system **INS (Initial navigation system)**, which understand how much the flight turns and banks using an gyroscope these track every single turn by the plane and indicates the location of the plane by changing the location from the initial reference which could have misguided if it failed to track every turn properly. The charts recovered from the debris is found to have wrong **minimum decent altitude** which is the height at which the pilots must be able to see the airport or must perform a missed approach if failed. Which means the crew were flying in a dangerous low altitude as per the US standards for such a terrain. The route was not approved and reviewed by the US department.

## Changes that were brought after the incident

CVR and FDR are installed in all airforce planes. No planes should be landed on airports that was no reviewed by the department of defense which includes for high ranking diplomats

## Reference

1. USAF Flight 21 Mystery — Mayday Air Disaster  
<https://www.youtube.com/watch?v=0fN4RjH0ByQ>